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Faunal and Floral Migrations and Evolution in SE Asia-Australasia

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Palaeozoic and Mesozoic tectonic evolution and biogeography of SE Asia-Australasia

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ABSTRACT: Southeast Asia* comprises a complex assembly of continental fragments (terranes), bounded by suture zones that represent former ocean basins. In the Early Palaeozoic (545-410 Ma), all the principal Southeast Asian continental terranes were located on the Himalayan-Australian Gondwanaland margin, where they formed part of a "Greater Gondwanaland". Multidisciplinary studies suggest that the Asian terranes rifted and separated from NW Australian Gondwanaland as three continental slivers around 350, 270, and 200-140 Ma. During the separation of these continental slivers, three ocean basins, the Palaeo-Tethys, Meso-Tethys and Ceno-Tethys, opened between each successive sliver and Gondwanaland. The first sliver, interpreted to have separated at 350 Ma, included North China, South China, Tarim, and Indochina. By 325 Ma, these terranes had no Gondwanaland faunas and floras, and palaeomagnetic data indicate that they were separated from the parent craton. Clockwise rotation of Gondwanaland around 320-300 Ma sent Australia into high southern latitudes, and glaciation affected much of Gondwanaland from 320 to 270 Ma. During this time, glacial ice reached the marine environment and glacial-marine sediments were deposited on the Sibumasu, Qiangtang, and Lhasa terranes now located in South and Southeast Asia. Gondwanaland cold-climate faunas and floras and cold water oxygen isotopic signatures are also found in Lower Permian rocks of these terranes. Rifting, starting at about 300 Ma, led to separation of the Sibumasu and Qiangtang terranes from Gondwanaland at approximately 270 Ma. These Cimmerian terranes drifted rapidly northwards between 270 and 250 Ma and significant changes in both brachiopod and fusulinid biogeographic provinces on these between 300 and 250 Ma are interpreted as being primarily due to the separation and northwards drift of the Cimmerian continental terranes, coupled with climatic amelioration following retreat of the Gondwanaland glaciation. Rifting and separation of a third continental sliver from Gondwanaland began 230 million years ago with the separation of the Lhasa block and progressed eastwards with separation of West Burma and possibly the small Sikuleh terrane of SW Sumatra and other small microcontinental fragments now located in Borneo and West Sulawesi around 160 Ma. Amalgamation and accretion of Gondwanaland-derived Asian terranes occurred progressively between the Late Devonian and Cretaceous.

1 INTRODUCTION

Faunal and floral links between SE Asia and Australasia go back more than 500 million years and the waxing and waning of biogeographic affinities between the two regions directly reflect the geological and tectonic histories of and interactions between these regions. Palaeozoic and Mesozoic biogeographic studies of the SE Asian region must be undertaken within a framework of the kinematic history of the allochthonous continental terranes of the region. Present-day distributions of animals and plants in SE Asia-Australasia are the result of both relatively recent dispersals, and more ancient vicariance events that can be related to the geological evolution of the region.

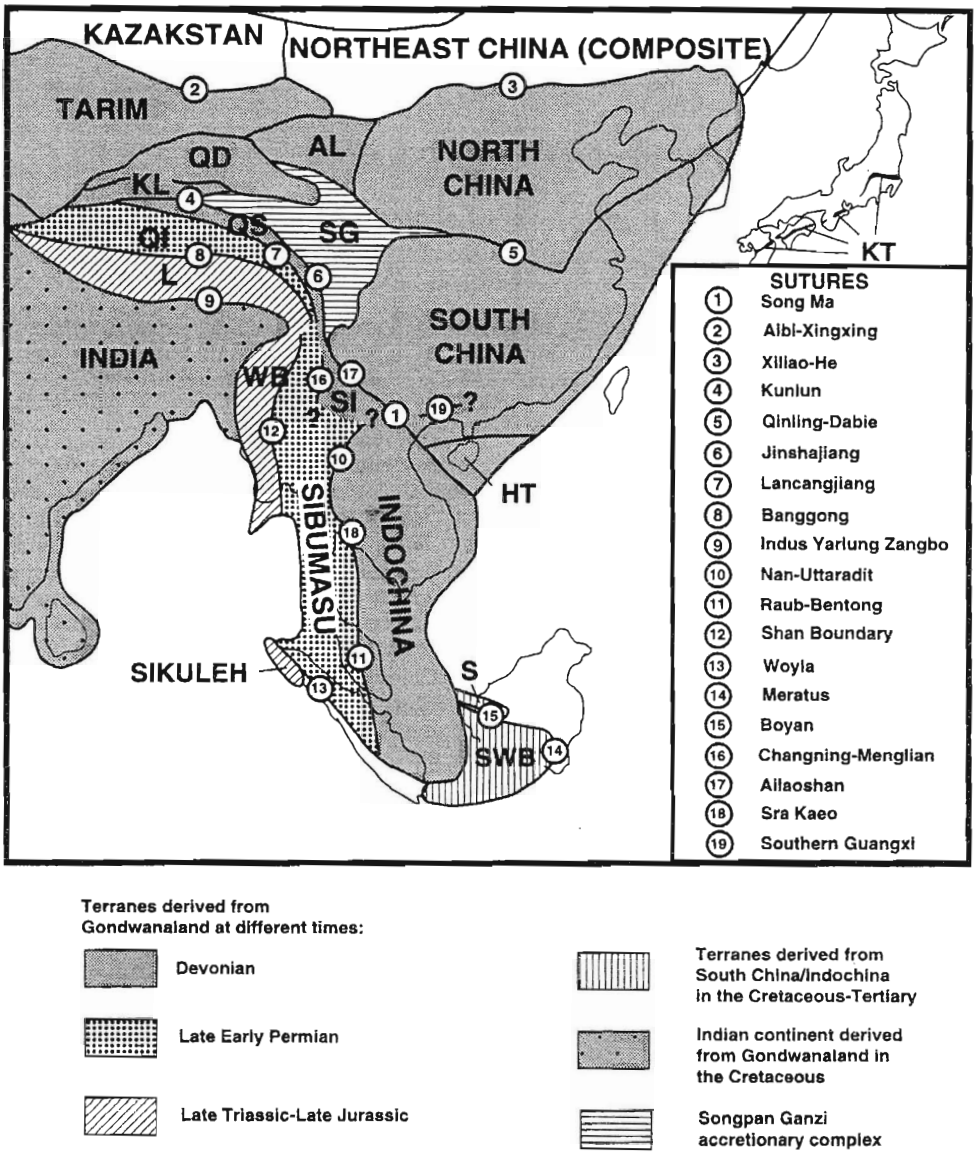


Figure 1. Distribution of principal continental terranes and sutures of East and Southeast Asia. WB = West Burma, SWB = South West Borneo, S = Semitau Terrane, HT = Hainan Island terranes, L = Lhasa Terrane, QI = Qiangtang Terrane, QS = Qamdo-Simao Terrane, SI= Simao Terrane, SG = Songpan Ganzi accretionary complex, KL = Kunlun Terrane, QD = Qaidam Terrane, AL = Ala Shan Terrane, KT = Kurosegawa Terrane.

East and Southeast Asia comprise a complex assembly of continental fragments (terranes), which are bounded by suture zones that represent the remnants or sites of former ocean basins (Figures 1 and 2). Biogeographical and other data suggest that in the Early Palaeozoic (545-410 Ma) all of the principal East and Southeast Asian continental terranes were located on the India-Australian margin of the ancient supercontinent Gondwanaland, where they formed part of a "Greater Gondwanaland" (Metcalfe, 1988, 1990, 1993, 1996, 1998, 1999; Burrett *et al.* 1990).

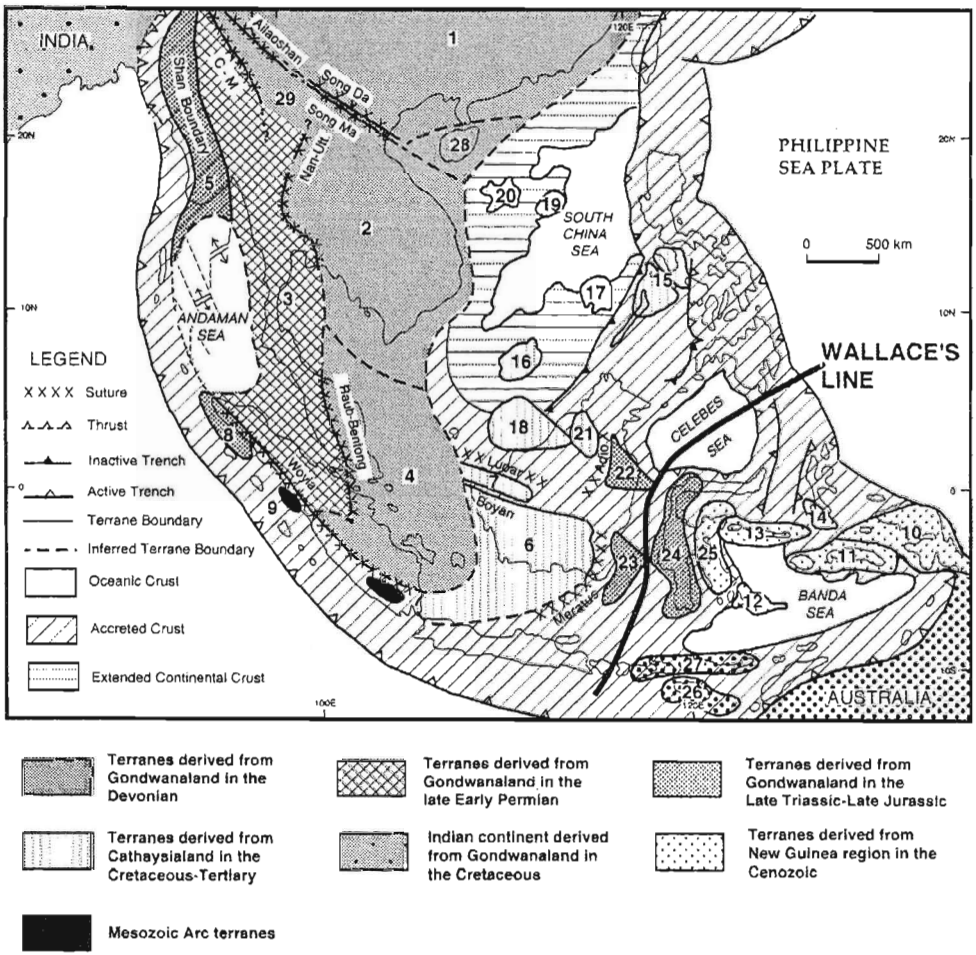


Figure 2. Distribution of continental blocks and fragments (terranes) and principle sutures of Southeast Asia in relation to Wallace's Line (m odified after Metcalfe, 1990). 1. South China 2. North Indochina 3. Sibumasu 4. Hainan Island terranes 5. West Burma 6. S.W. Borneo 7. Semitau 8. Sikuleh 9. Natal 10. West Irian Jaya 11. Buru-Seram 12. Buton 13. Bangai-Sula 14. Obi-Bacan 15. North Palawan 16. Spratley Islands-Dangerous Ground 17. Reed Bank 18. Luconia 19. Macclesfield Bank 20. Paracel Islands 21. Kelabit-Longbowan 22. Mangkalihat 23. Paternoster 24. West Sulawesi 25. East Sulawesi 26. Sumba 27. Qamdo-Simao terrane.

Multidisciplinary data (e.g., biogeographic, biostratigraphic, palaeomagnetic, structural/tectonic) suggest that the Asian terranes rifted and separated from NW Australian Gondwanaland as three continental slivers around 350, 270, and 200-140 million years ago, in the Devonian, Early Permian, and Late Triassic-Late Jurassic respectively. During the separation of these continental slivers, three ocean basins, the Palaeo-Tethys, Meso-Tethys, and Ceno-Tethys, are interpreted to have opened between each successive sliver and Australian Gondwanaland. Closure of these ocean basins by subduction processes led to the juxtaposition of once widely separated continental fragments, and the remnants of the ocean basins are now preserved in the suture zones of the region (Figure 3).

During this complex evolution, reorganisation of continent-ocean configurations, the formation of the supercontinent Pangea, and its subsequent breakup, coupled with major global cli-

matic shifts, led to dramatic changes in the biogeography of organisms. This paper discusses the tectonic evolution of the SE Asian region in relation to Australasia, and the changing SE Asian biogeographic patterns observed in the Palaeozoic and Mesozoic that are related to this tectonic evolution.

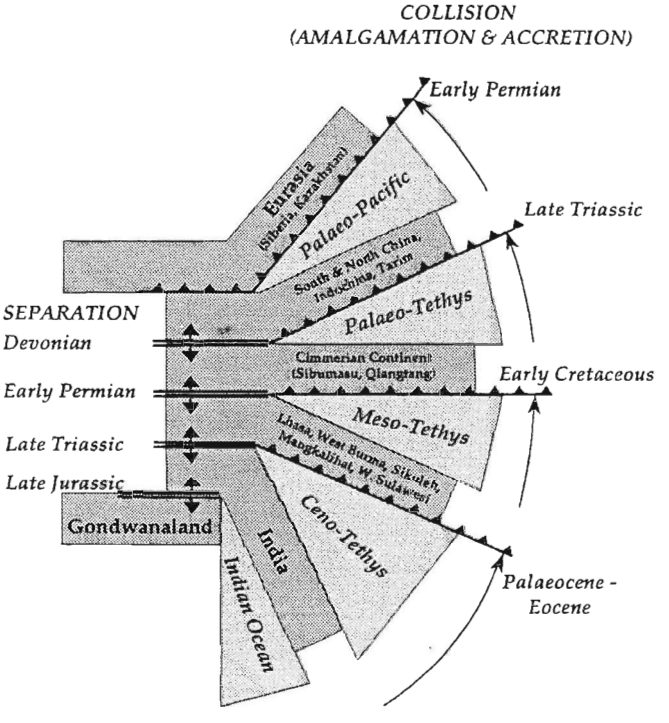


Figure 3. Schematic diagram showing times of separation and subsequent collision of the three continental slivers/collages of terranes and India that rifted from Gondwanaland and translated northwards by the opening and closing of three successive oceans, the Palaeo-Tethys, Meso-Tethys and Ceno-Tethys.

2 PALAEOZOIC AND MESOZOIC TECTONIC EVOLUTION AND BIOGEOGRAPHY

The East and Southeast Asian terranes represent three collages that successively rifted and separated from the India-Australian margin of Gondwanaland as three elongate continental slivers. As these three slivers separated from Gondwanaland in the Devonian, late Early Permian and Late Triassic-Late Jurassic, three ocean basins, the Palaeo-Tethys, Meso-Tethys and Ceno-Tethys were opened and subsequently closed (Metcalfe, 1990, 1996, 1998, 1999). Australia remained part of eastern Gondwanaland (attached to Antarctica and India) during the Palaeozoic and early Mesozoic and finally separated from Antarctica at about 45 Ma, after which it drifted northwards eventually to begin its currently ongoing collision with SE Asia. The principal tectonic events and the changing biogeography of organisms in the SE Asia-Australasian region during the Palaeozoic and Mesozoic are outlined below:

2.1 *Cambrian-Ordovician-Silurian (545 - 410 Ma)*

Close faunal affinities between the East and Southeast Asian terranes and Australia are seen in the Cambrian, Ordovician, and Silurian (Burrett, 1973; Burrett and Stait, 1985; Metcalfe, 1988, Burrett *et al.*, 1990), and distinctive Asian-Australian faunal provinces can be recognised during

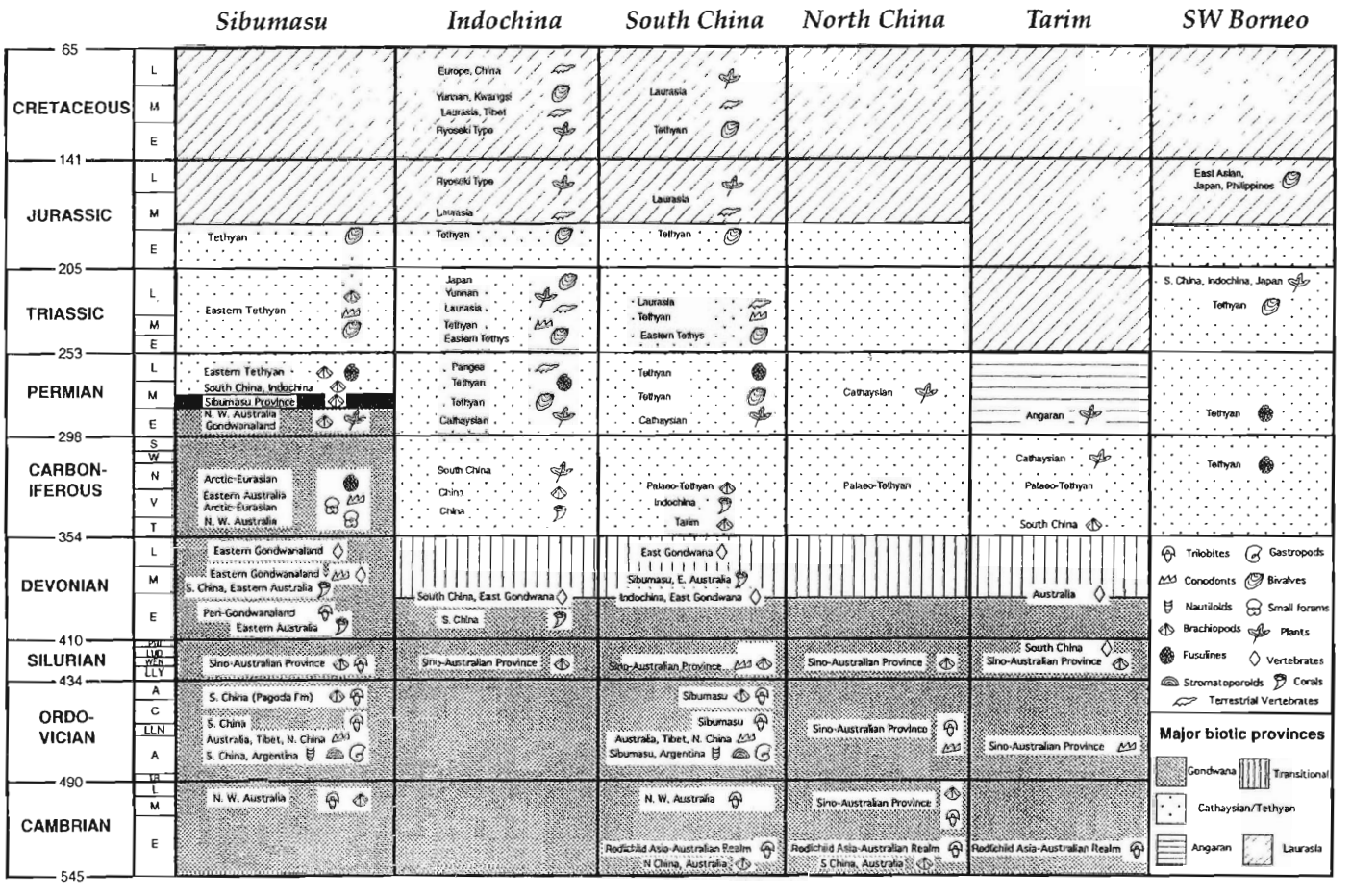


Figure 4. Palaeozoic and Mesozoic faunal and floral provinces and faunal/floral affinities vs. time for the principal East Asian continental terranes. For explanation see text.

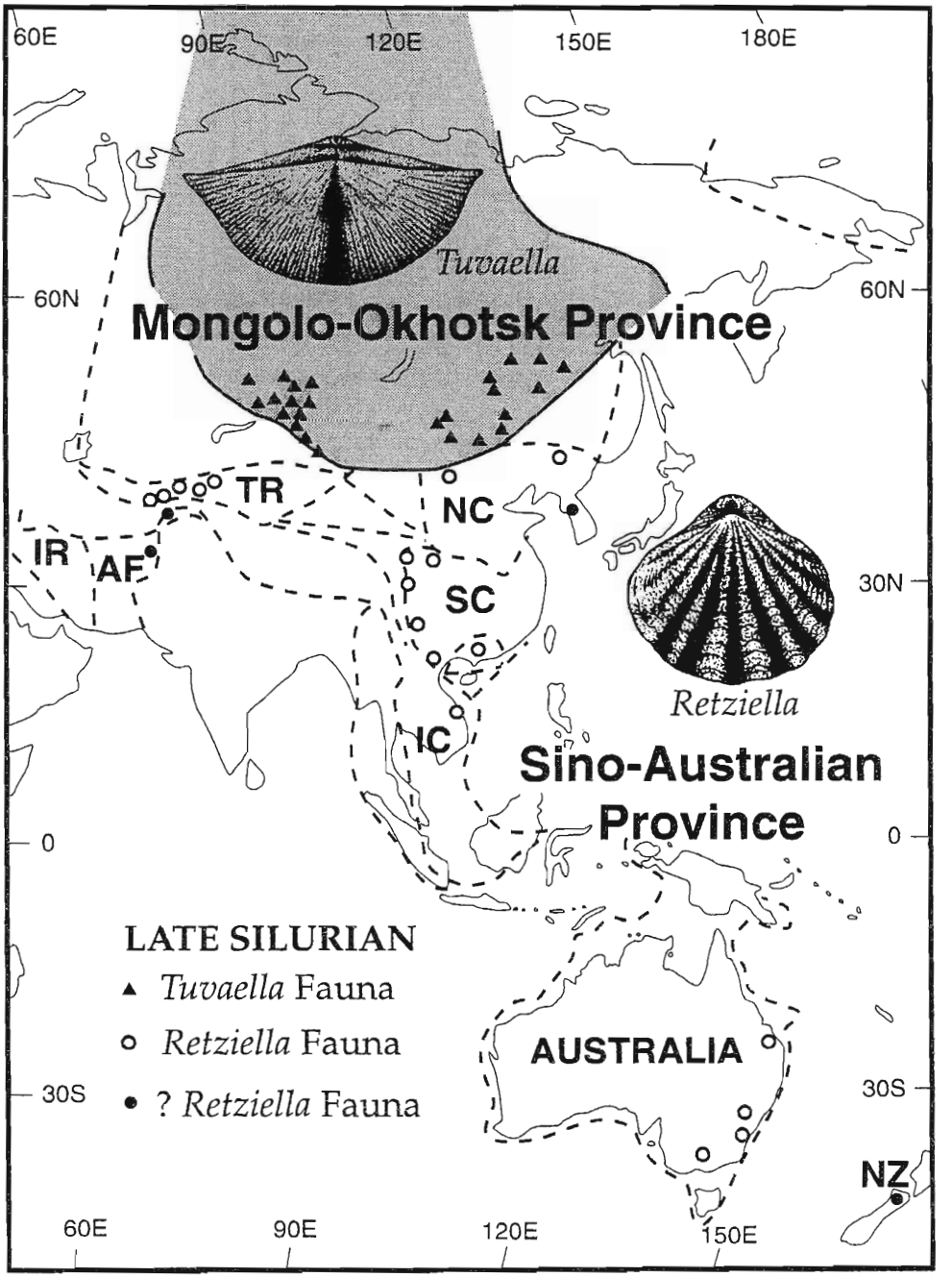


Figure 5. Late Silurian brachiopod provinces of Asia-Australasia (After Rong *et al.* 1995).

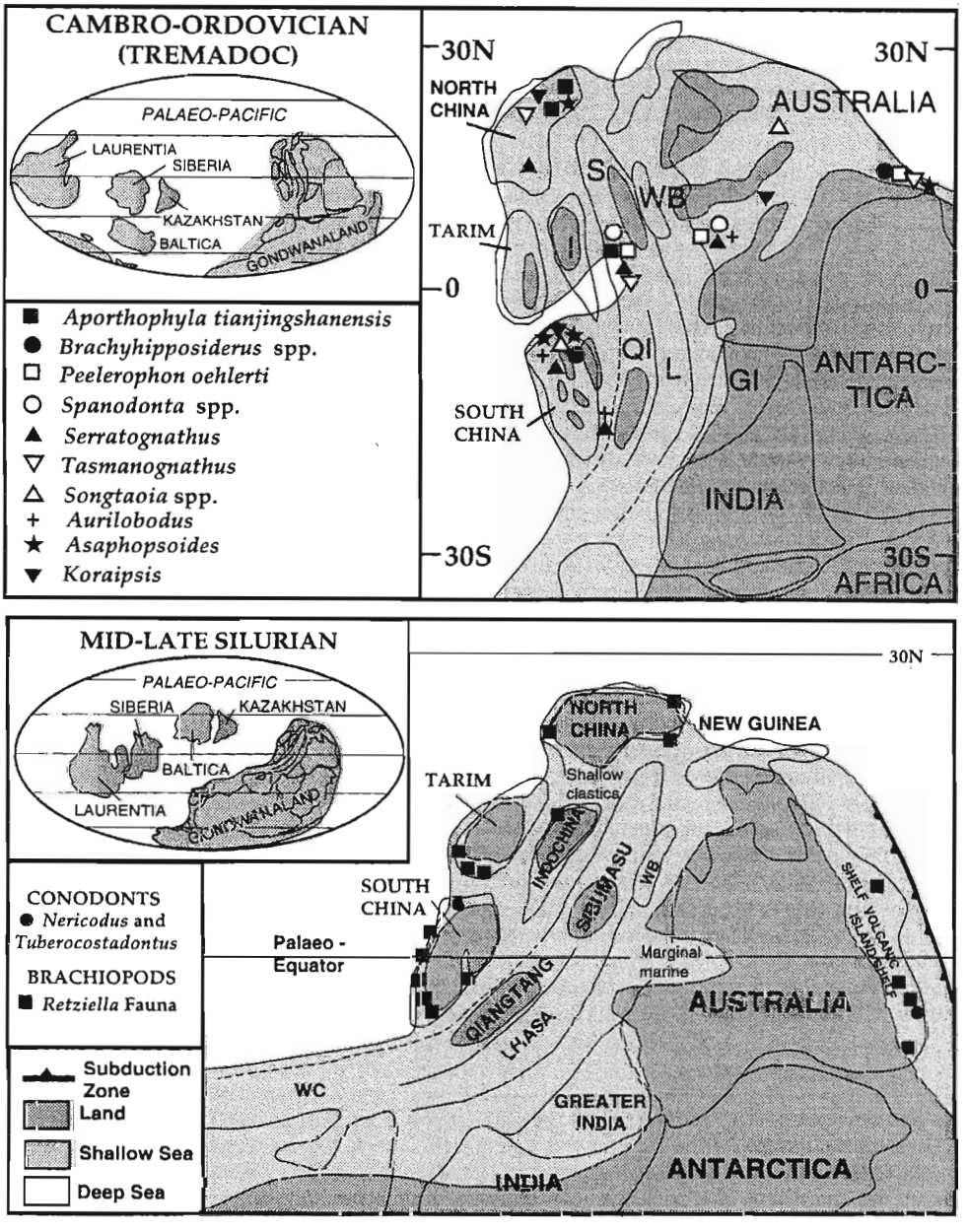


Figure 6. Reconstructions of eastern Gondwanaland for the Cambro-Ordovician (Tremadoc) and Mid-Late Silurian showing the postulated positions of the East and Southeast Asian terranes, distribution of land and sea, and shallow-marine fossils that illustrate Asia-Australia connections at these times. I = Indochina QI = Qiangtang L = Lhasa S = Sibumasu WB = West Burma WC = Western Cimmerian Continent GI = Greater India. Present day outlines are for reference only. Distribution of land and sea for Chinese blocks principally from Wang (1985). Land and sea distribution for Pangea/Gondwanaland compiled from Golongka *et al.* (1994), Smith *et al.* (1994); and for Australia from Struckmeyer & Totterdell (1990).

this time interval (Figure 4). During the Cambrian, trilobite faunas of Asia-Australasia define an Asian-Australian Faunal Realm (Yang, 1994) and archaeocyaths and in particular lingulate brachiopods suggest close proximity of both North and South China to Australasia in the Early Cambrian (Brock *et al.,* 1999). Late Cambrian trilobites (Shergold *et al.*, 1988) and brachiopods (Yang, 1994) of Sibumasu, South and North China, and Australia define a Sino-Australian province. Ordovician trilobites (Burrett *et al.*, 1990; Fortey, 1997; Fortey and Cocks, 1998), brachiopods (Laurie and Burrett, 1992), corals and stromatoporoids (Webby *et al.*, 1985; Lin and Webby, 1989), nautiloids (Stait and Burrett, 1982, 1984; Stait *et al.*, 1987), gastropods (Jell *et al.*, 1984), and conodonts (Burrett *et al.*, 1990; Nicoll and Totterdell, 1990; Nicoll and Metcalfe, 1994, 1999) again define a Sino-Australian province. Early Ordovician trilobites from Dali, West Yunnan, on the Qamdo-Simao terrane are regarded as Gondwanan cold-water forms by Zhou *et al.* (1998a) with close relatives in south-central Europe and the Yangtze region of South China. Ordovician (Late Areng-early Llanvirn) trilobites from the Tarim terrane indicate that Tarim and South China were close to each other in Ordovician times, with 80% forms common between the Xinjiang region of the Tarim terrane and the Yangtze region of South China. The presence of typical Gondwana forms suggests these terranes were peri-Gondwana in the Ordovician (Zhou *et al.* 1998b). Upper Ordovician (Caradoc) trilobite faunas from southern Thailand on the Sibumasu terrane, dominated by *Ovalocephalus*, nileids and remopleuridids, are identical, even at the species level to faunas described from the Pagoda Limestone of South China (Fortey, 1997), indicating close proximity of these two terranes in the Late Ordovician. In addition, Upper Ordovician (Ashgill) to Lower Silurian (Llandovery) brachiopods and trilobites from south Thailand and the Northern Shan States of Burma on the Sibumasu terrane are identical to those reported from the Yangtze region of South China (Cocks and Fortey 1997). Furthermore, the sequence of faunas and stratigraphy from at least the Middle Ordovician to the Middle Silurian of the Thai-Burma part of Sibumasu and the Yichang area of South China was said by Cocks and Fortey (1997) to be so similar that these regions must have been adjacent to one another during this time period. In addition, the Gondwanaland acritarch *Dicrodiacroium ancoriforme* Burmann has been reported from the Lower Ordovician of South China (Servais *et al.*, 1996). Cambrian-Ordovician faunas of Indochina are poorly known and their biogeographic affinities are not clear. However, Silurian brachiopods of Indochina, along with those of South China, North China, Eastern Australia, and the Tarim terrane, belong to the Sino-Australian province (Figure 5) characterised by the *Retziella* fauna (Rong *et al.*, 1995). In addition, Silurian conodonts provide links between South China and Australia (Nicoll and Metcalfe, 1994; Nicoll and Metcalfe,1999). These biogeographic links in the Early Palaeozoic, together with similar stratigraphies (Metcalfe, 1996; Fortey and Cocks, 1998) and palaeomagnetic data (Zhao *et al.*, 1996; Klootwijk, C. 1996) suggests that the principal East and Southeast Asian terranes (Sibumasu, Indochina, South China, North China and Tarim) were positioned adjacent to the Himalayan-NW Australian margin of Gondwanaland forming part of a "Greater Gondwanaland" which was located in low equatorial latitudes during the early Palaeozoic (Figure 6).

2.2 *Devonian (410 - 354 Ma)*

The Devonian period was one of biogeographic transition for the principal East and Southeast Asian terranes, except for the Sibumasu terrane which continued to be biogeographically part of Gondwanaland until the early Sakmarian (Early Permian). Early Palaeozoic Gondwanaland faunas of the Indochina, South China, North China, and Tarim terranes give way to Cathaysian/Tethyan floras and faunas in the Carboniferous (Figure 4). Devonian faunas and floras of these terranes appear to have mixed biogeographical affinities. Lower Devonian (Emsian) trilobites from Thailand on the Sibumasu terrane can be closely compared to those of the same age in Turkey, Bohemia, Germany, and Morocco, and they belong to the peri-Gondwana "Hercynian magnafacies" (Fortey, 1989). Middle Devonian rugose corals of Sibumasu have close species affinities with those of South China and eastern Australia (Aung, 1995) and Middle Devonian polygnathid conodonts of Sibumasu also indicate connections to South China and eastern Gondwanaland (Burrett *et al.*, 1990). Of particular interest is Devonian vertebrate biogeography and especially that of shallow-marine and fresh-water fish. Much debate has raged in recent years regarding late Palaeozoic global reconstructions based on biogeography vs. palaeomagnetism, with some apparent conflicts between vertebrate biogeography and palaeomagnetism. An ex-

ample of this is the "Amorica" problem as discussed by Young (1987) where Devonian vertebrate biogeography suggests that Gondwana and Euramerica were contiguous by Middle Devonian times, in contrast to palaeogeographic reconstructions based on palaeomagnetic data which imply a wide ocean separating these regions during the Devonian. Alternative explanations of the biogeography involve a continental block "Amorica" which traversed the ocean during the Devonian. Provincialism of Devonian vertebrates has long been known, and Young (1981) recognised five major vertebrate provinces for Early Devonian times (Figure 7A). At that time, the Devonian vertebrate faunal affinities of the North China, Indochina, Tarim, and Sibumasu terranes were largely unknown. Since then, much new information has come to light and was recently summarised by Young and Janvier (1999). It is now known that Middle to Late Devonian fish of Sibumasu (e.g., the thelodont *Turinia pagoda* Wang, Deng & Turner) have strong affinities with closely allied forms in the Officer Basin of Australia (Long, 1993) and also similarities with fish faunas of South China and east Gondwanaland (Long and Burrett 1989). It is also now known that galeaspids and sinolepids occur on North China, but interestingly, are somewhat younger than the South China forms. South China Devonian "endemic" fish are also now known to occur on the Tarim block and on Indochina (Tong-Dzuy and Cai 1995; Tong-Dzuy *et al.* 1996; Young and Janvier 1999). New discoveries of Devonian fish faunas linking the principal Asian terranes led Young and Janvier to propose that these terranes formed an "Asian Superterrane" in the Middle Palaeozoic and suggested that this superterrane was either located off NW Australia or more likely (in their view) outboard of eastern Australia (Figure 7B). The configuration of Young and Janvier's "Asian Superterrane" presents Tarim, North China, South China, and Indochina in their current relative positions. This is untenable in view of our extensive knowledge from the suture zones between these terranes indicating that the Qinling-Dabei suture zone segment of Palaeo-Tethys between North and South China did not close until Late Permian-Triassic times! Similarly, suturing between Indochina and South China probably occurred in the early Carboniferous, and the North China and Tarim terranes could not have had their current relative positions in the Middle Palaeozoic based on palaeomagnetic and other data. However, the positioning of the Tarim, North and South China, and Indochina terranes off NW Australia, but with rather different relative positions, is consistent with a range of multidisciplinary data (Metcalfe, 1996, 1998). Young and Janvier, however, considered that this position, "within the palaeo-Tethys" (sic) would not allow for sufficient isolation to develop the observed endemic continental fish faunas. A position off eastern Australia (position B in Figure 7B), was therefore favoured; this leans heavily on the controversial position of South China between Eastern Australia and Laurentia in a Late Proterozoic reconstruction of Rodinia (Li *et al.* 1995, 1996). Such a position in the Middle Palaeozoic would be precluded by our knowledge of the early Palaeozoic biogeography of the Asian terranes (see above) and the geology of eastern Australia which implies westwards directed subduction of the "Palaeo-Pacific" beneath eastern Australia. That would have resulted in the "Asian superterrne" being accreted to eastern Australia! I therefore prefer to position these terranes (though not in their current relative positions) off NW Australian Gondwana in the Devonian (Figure 8), and explain the waxing and waning biogeographic patterns observed on and between these terranes and Gondwanaland during the Middle to Late Palaeozoic by the process of rifting and separation of these terranes from Gondwanaland, their northwards drift, and the opening of the Palaeo-Tethys. Early Devonian endemic continental fish faunas could have developed because of shifting distributions of land and sea and the development of intracontinental marine barriers during these processes and are not necessarily the result of continental isolation in a true tectonic sense.

East Australia continued to reside in low southern latitudes during the Devonian but rotated counterclockwise. This counterclockwise rotation was concurrent with the clockwise rotation of the Chinese terranes as they separated from Gondwanaland as an elongate continental sliver. As this sliver separated from Gondwanaland, the Palaeo-Tethys ocean opened up (Figure 8). Late Devonian faunas on the Chinese terranes still have some Australian connections which are well illustrated by the occurrence of the antiarch (placoderm) group Sinolepidoidei (including *Grenfellaspis*) described exclusively from Grenfell in southeast Australia and from the North and South China terranes (Ritchie *et al.* 1992; Johanson and Ritchie 1999).

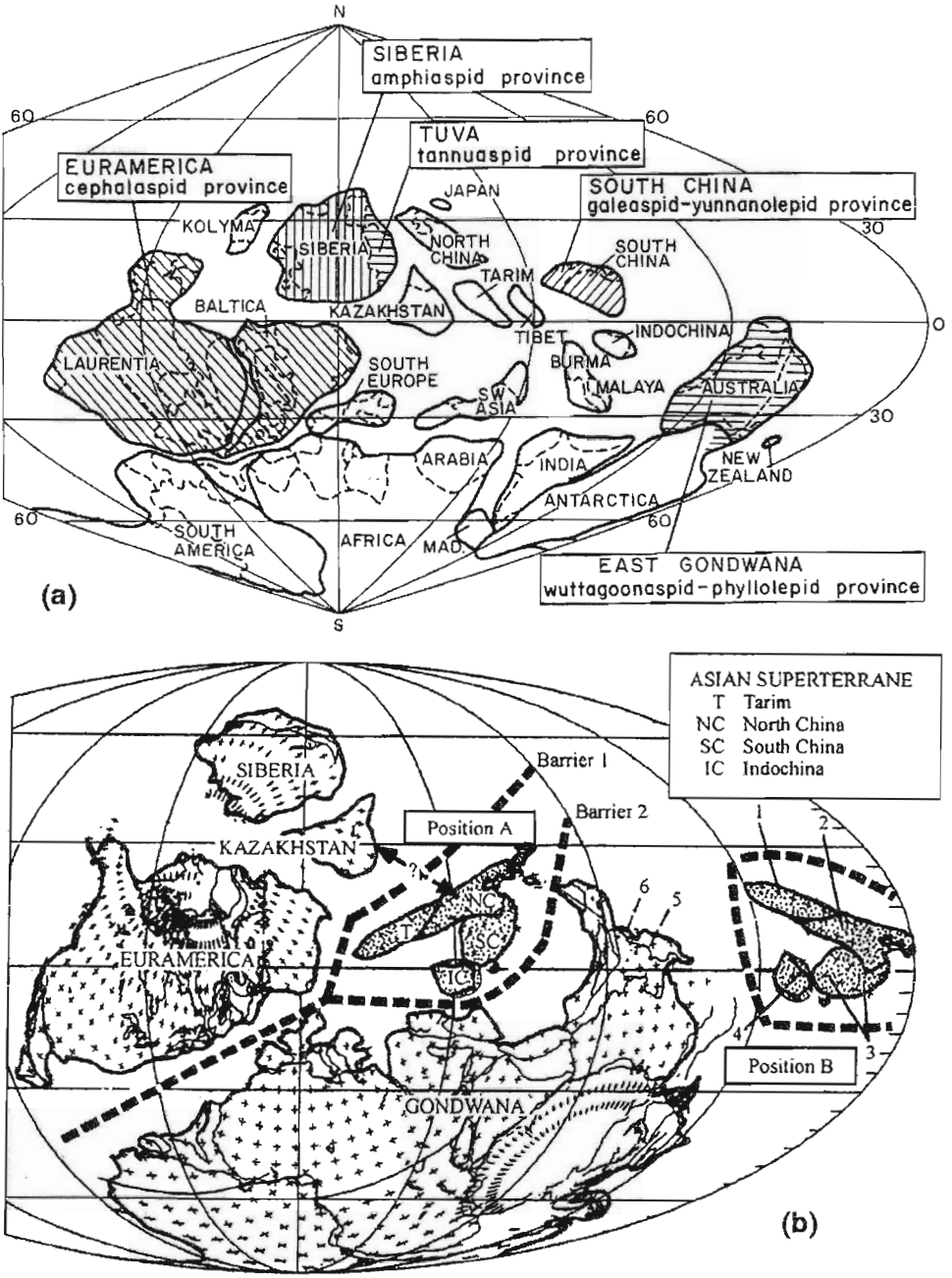


Figure 7. A. The five Devonian fish provinces of Young (1981). B. Alternative Late Palaeozoic positions proposed for the "Asian Superterrane" by Young and Janvier (1999).

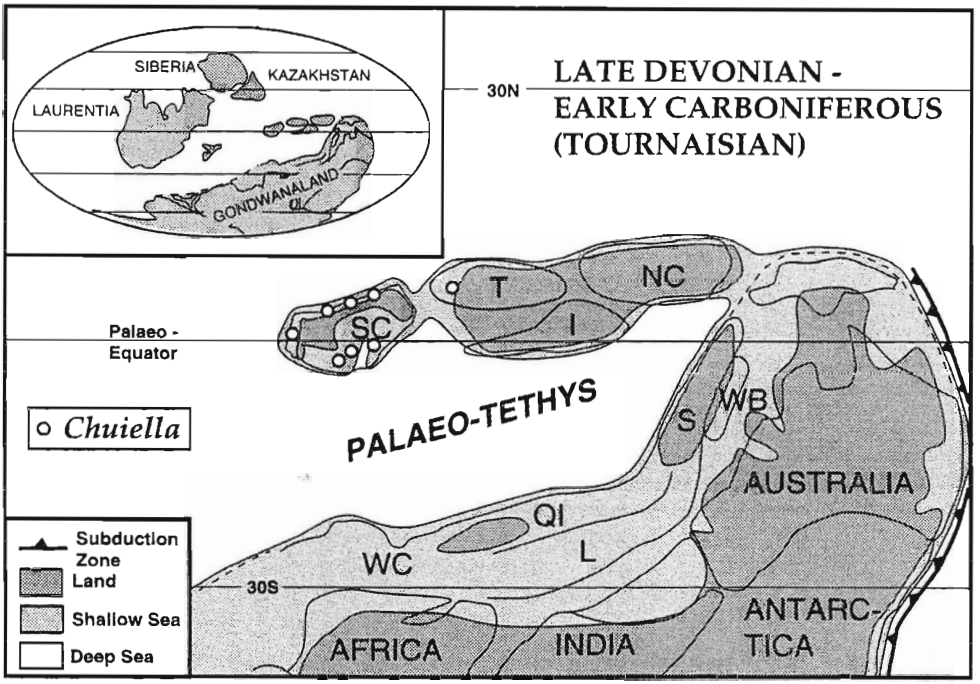


Figure 8. Reconstruction of eastern Gondwanaland for the Late Devonian to Lower Carboniferous (Tournaisian) showing the postulated positions of the East and Southeast Asian terranes, distribution of land and sea, and opening of the Palaeo-Tethys ocean at this time. Also shown is the distribution of the endemic Tournaisian brachiopod genus *Chuiella* (from Chen and Shi 1999). Present-day outlines are for reference only. Distribution of land and sea for Chinese blocks principally from Wang (1985). Land and sea distribution for Pangea/Gondwanaland compiled from Golongka *et al.* (1994), Smith *et al.* (1994); and for Australia from Struckmeyer & Totterdell (1990). Symbols as for figure 6.

2.3 *Carboniferous (354 - 298 Ma)*

During the Carboniferous, Gondwanaland rotated clockwise and collided with Laurentia in the west to form Pangea. Australia, still attached to northeast Gondwanaland, drifted from low southern latitudes (0° to 40°S) in Tournaisian - Visean times to high southern latitudes (50° to 75°S) in middle-late Namurian times. The major Gondwanan glaciation commenced in the Namurian and extended through into the Early Permian. There were major global shifts in both plate configurations and climate during this time and a change from warm to cold conditions in Australasia. This is reflected in the change from high- to low-diversity faunas in Australasia (Jones *et al.*, 2000) and especially eastern Australia where low diversity, endemic faunas developed in the Upper Carboniferous (Figure 9). Carboniferous faunas and floras in North and South China, Indochina, and Tarim are tropical/sub-tropical Cathaysian/Tethyan types and show no Gondwanaland affinities. These terranes had already separated from Gondwanaland and were located in low-latitude/equatorial positions during the Carboniferous (Figure 9). Proximity of Tarim and South China in the Tournaisian is indicated by common endemic species of the brachiopod genus *Chuiella* (Chen and Shi 1999; Figure 8). Indochina and South China collided within the Tethys during the early Carboniferous and amalgamated along the Song Ma suture zone, now located in Laos and Vietnam. Late Lower Carboniferous (uppermost Mississippian) floras on these terranes are similar and suggest continental connection at this time (Laveine *et al.* 1999). Ice sheets and glaciers extended across much of eastern Gondwanaland during the Late Carboniferous.

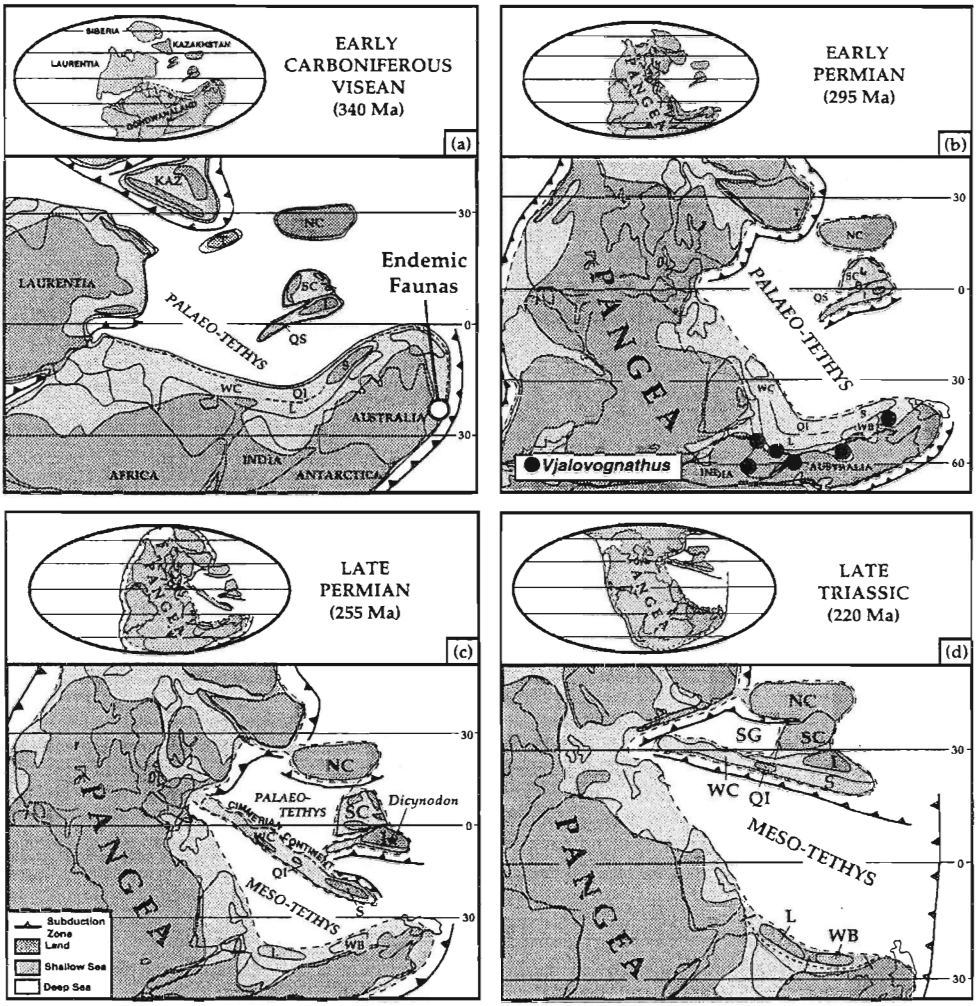


Figure 9. Palaeogeographic reconstructions of the Tethyan region for (a) Early Carboniferous (Visean), (b) Early Permian, (c) Late Permian, and (d) Late Triassic showing relative positions of the East and Southeast Asian terranes and distribution of land and sea. The distribution of the Lower Permian cold-water-tolerant conodont genus *Vjalovognathus* and the location of the Late Permian *Dicynodon* from Laos are also shown. Present day outlines are for reference only. Distribution of land and sea for Chinese blocks principally from Wang (1985). Land and sea distribution for Pangea/Gondwanaland compiled from Golonka *et al.* (1994), Smith *et al.* (1994); and for Australia from Struckmeyer & Totterdell (1990). Symbols as for figure 6.

2.4 *Early Permian (298 - 270 Ma)*

During the Permian, Australia remained in high southern latitudes. Glacial ice reached the marine environment of the northeast Gondwanaland margin and glacial-marine sediments were deposited on the Sibumasu, Qiangtang, and Lhasa terranes. Gondwanaland cold-climate faunas and floras characterised the Sibumasu, Qiangtang, and Lhasa terranes at this time. In addition, the distinctive cool-water tolerant conodont genus *Vjalovognathus* defines an eastern peri-Gondwanaland cold-water province (Nicoll and Metcalfe, 1998; Figure 9). Early Permian (Asselian-Early Sakmarian) brachiopod faunas of the North China, South China, Tarim, and Indochina terranes defined a Cathaysian province, whilst coeval brachiopod faunas of the Sibumasu,

Qiangtang, and Lhasa terranes formed part of the peri-Gondwana Indoralian Province (Shi and Archbold, 1998). The biogeographic contrast seen in the Lower Permian between terranes south west and northeast of the Palaeo-Tethyan Lancangjian, Changning-Menglian, Nan-Uttaradit, Sra Kaeo, and Bentong-Raub suture zones (Figure 10) is as striking as, if not more striking than that seen either side of Wallace's Line. This remarkable juxtaposition of Lower Permian cold-climate Gondwanaland fauanas and floras with warm-climate, equatorial Cathaysian faunas and floras is the result of the closure of the Palaeo-Tethys ocean which had separated them by thousands of kilometers. This is graphically illustrated by the distribution of Lower Permian plants in the region (Figure 11) in which three floral provinces are recognised, the southern hemisphere cool-climate Gondwana province (on India and the Lhasa, Qiangtang, and Sibumasu terranes), the palaeoequatorial warm, subtropical to tropical Cathaysian province (on North China, South China, and Indochina) and the northern hemisphere cool-climate Angaran province (on the Tarim terrane, Kazakstan, and Northeast China). Plotting of these floral provinces on a Lower Permian reconstruction of Pangea (Figure 11B) brings these floras into appropriate latitudinal positions and also demonstrates the relative isolation of the low latitude Cathaysian flora on the intra-Tethyan North China, South China, and Indochina terranes.

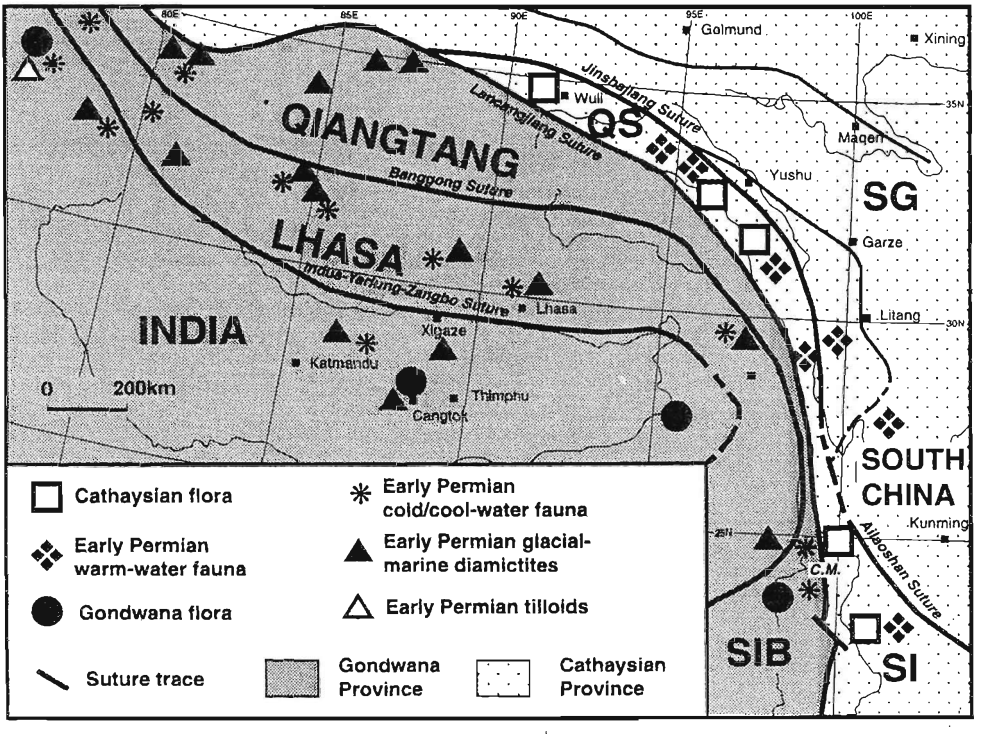


Figure 10. Distribution of Lower Permian Gondwana and Cathaysian province faunas and floras of the Tibet-Yunnan region, showing the remarkable juxtaposition of these highly contrasting cool- and warm-climate biotas either side of the main Palaeo-Tethyan divide represented by the Lancangjian and Changning-Menglian (C.M.) suture zones. QS = Qamdo-Simao terrane, SIB = Sibumasu terrane, SI = Simao terrane, SG = Songpan Ganzi accretionary complex.

2.5 *Late Permian (270 - 252 Ma)*

During the late Early Permian to Late Permian (Late Sakmarian-Midian), the Cimmerian continental sliver (Sengor *et al.*, 1988), including the Sibumasu and Qiangtang terranes, rifted away from the northeastern margin of Gondwanaland, and by the early Late Permian the Sibumasu and Qiangtang terranes had separated from Gondwanaland and the Meso-Tethys opened

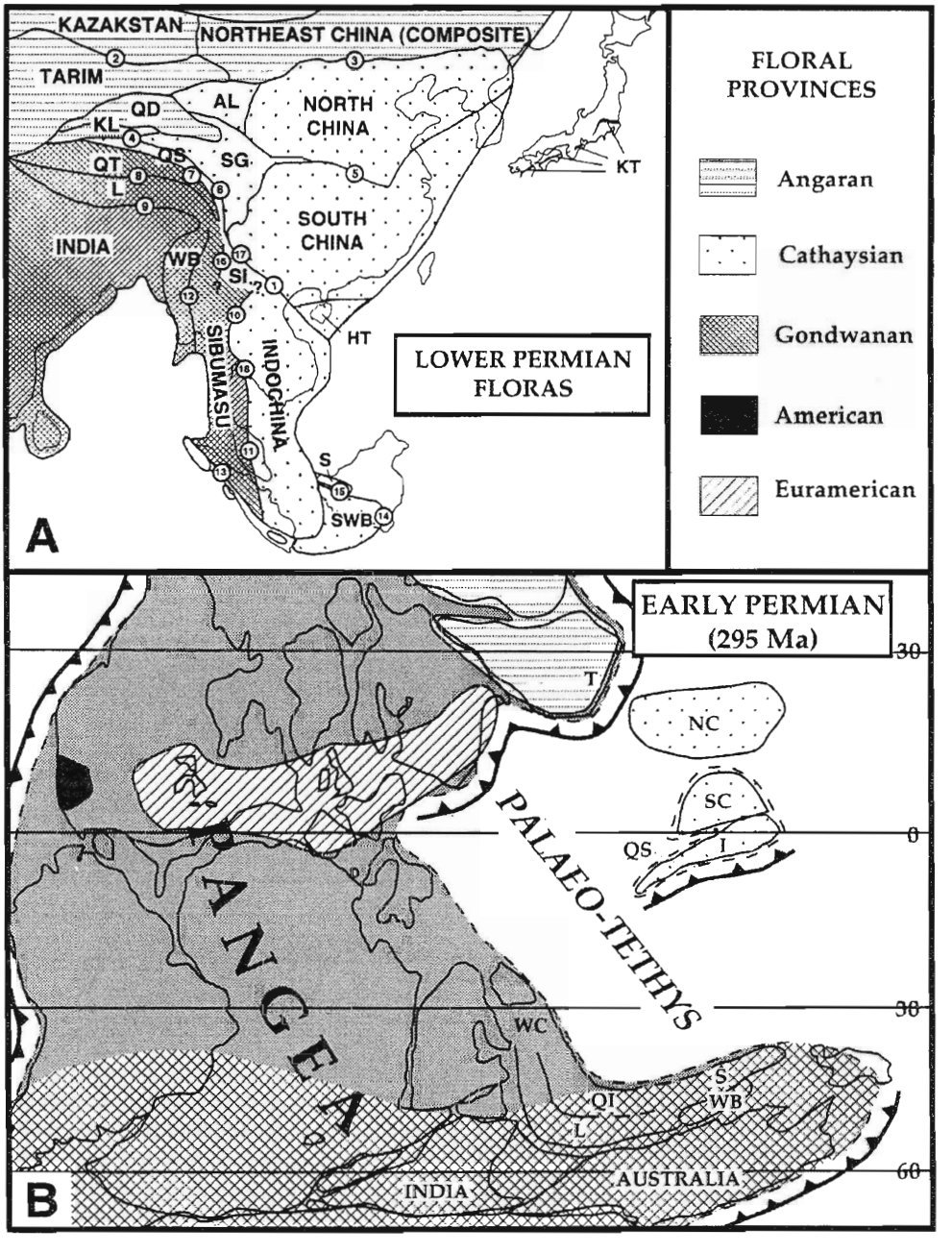


Figure 11. A. Distribution of Lower Permian floras on East and Southeast Asia terranes. B. Early Permian reconstruction of Pangea and Tethys showing positions of East and Southeast Asian terranes and floral provinces. WB = West Burma, SWB = South West Borneo, S = Semitau Terrane, HT = Hainan Island terranes, WC = Western Cimmerian Continent, L = Lhasa Terrane, QI = Qiangtang Terrane, QS = Qamdo-Simao Terrane, SI= Simao Terrane, SG = Songpan Ganzi accretionary complex, KL = Kunlun Terrane, QD = Qaidam Terrane, AL = Ala Shan Terrane, KT = Kurosegawa Terrane, T = Tarim Terrane, NC = North China, SC = South China, I = Indochina. Suture numbers as for Fig.1.

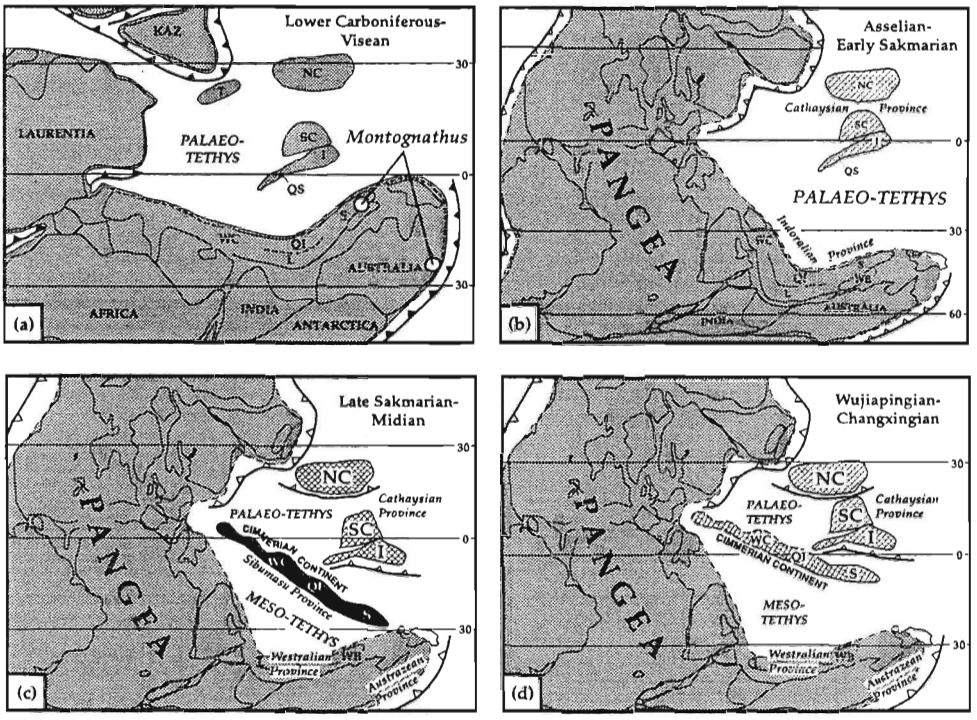


Figure 12. Palaeogeographic reconstructions for the (a) Lower Carboniferous (Visean) showing the occurrence of the eastern Gondwanaland endemic conodont genus *Montognathus*; and (b) Asselian-Early Sakmarian, (c) Late Sakmarian-Midian and (d) Wujiapingian-Changxingian, illustrating a tectonic vicariant model interpreting the change in marine provinciality of the Sibumasu and other elements of the Cimmerian continent during the Permian. Note that as Sibumasu separates from Gondwanaland and drifts northwards it loses its Indoralian (Gondwanaland) Province faunas, then develops endemic faunas representing an independent Sibumasu province, and finally becomes assimilated into the intra-Tethyan Cathaysian Province. After Shi and Archbold (1998).

between this continental sliver and Gondwanaland (Figure 9). The Palaeo-Tethys continued to be destroyed by northwards subduction beneath Laurasia, North China, and the amalgamated South China and Indochina terranes. During their northwards drift, after separation from Australian Gondwana, the Sibumasu and Qiangtang terranes developed initially a Sibumasu Province fauna and then were absorbed into the palaeoequatorial Cathaysian Province (Figure 12). This is particularly seen in the brachiopod faunas, but also in the fusulinid faunas, which are initially low-diversity cooler climate faunas characterised by *Monodiexodina* and *Eopolydiexodina* but are then replaced by higher diversity typical Tethyan fusulinid assemblages (Ueno, 1999). Brachiopod faunas of Australasia during Middle to Upper Permian times represent two provinces, a western Westralian Province and an eastern Austrazean Province (Archbold, 1996; Shi and Archbold, 1998). North and South China begin to collide during the Late Permian, and a connection between mainland Pangea and Indochina, via South and North China or via the western part of the Cimmerian continent, is indicated by the occurrence of the terrestrial vertebrate genus *Dicynodon* in the Upper Permian of Indochina (Figure 9).

2.6 *Triassic (252 - 205 Ma)*

Australia was in low to moderate southern latitudes during the Triassic. The Sibumasu and Qiangtang terranes collided and sutured to the Indochina/South China amalgamated terrane along the Lancangjian, Changning-Menglian, Nan-Uttaradit, and Bentong-Raub suture zones (Liu *et al.*, 1991, 1996; Metcalfe, 2000; Singharajwarapan and Berry, 2000).

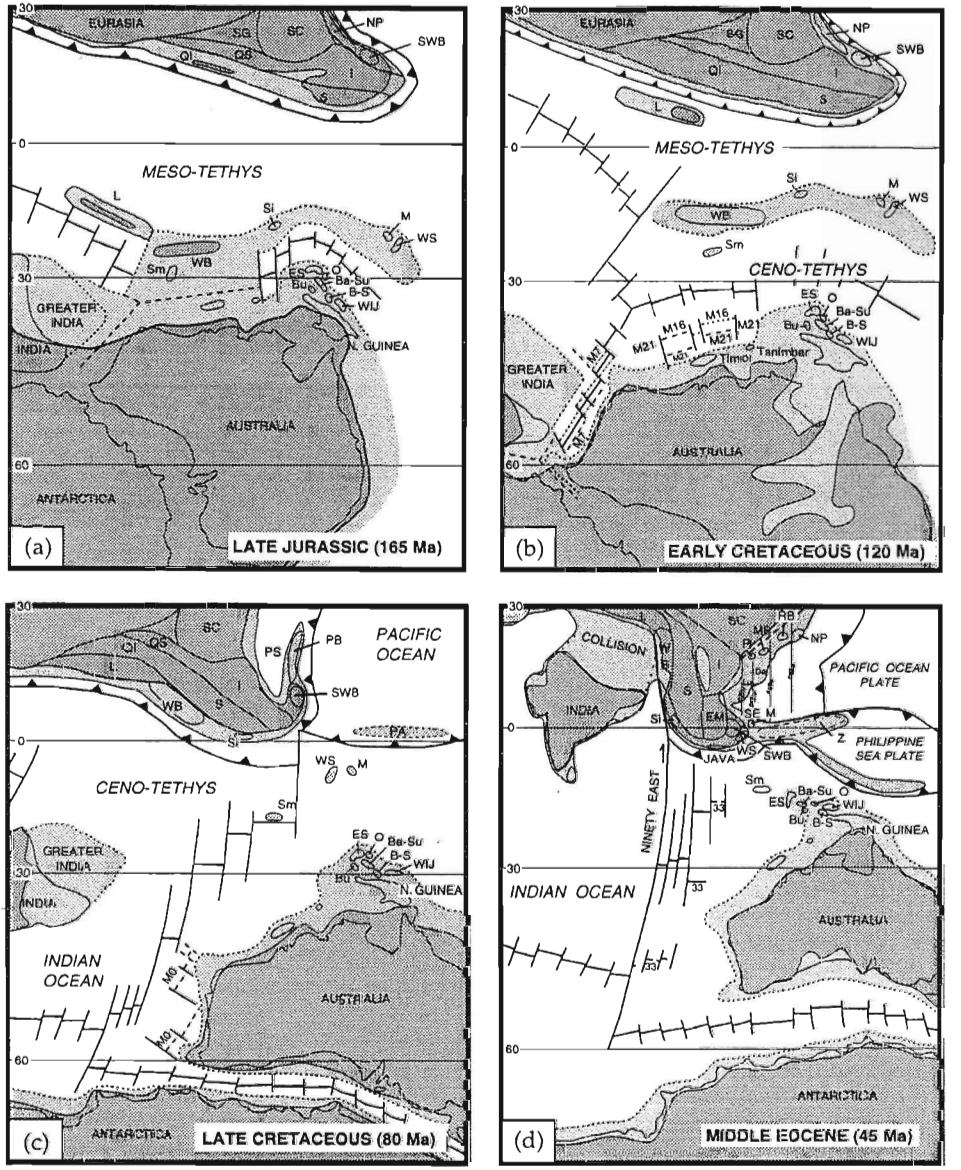


Figure 13. Palaeogeographic reconstructions for the East Asia-Australia region in (a) Late Jurassic, (b) Early Cretaceous, (c) Late Cretaceous, and (d) Middle Eocene times showing distribution of land and sea. SG = Songpan Ganzi accretionary complex SWB = South West Borneo (includes Semitau) NP = North Palawan and other small continental fragments now forming part of the Philippines basement Si = Sikuleh M = Mangkalihat WS = West Sulawesi ES = East Sulawesi O = Obi-Bacan Ba-Su = Bangai-Sula Bu = Buton B-S = Buru-Seram WIJ = West Irian Jaya Sm = Sumba PA = Insipient Philippine Arc. M numbers represent Indian Ocean magnetic anomalies. Other terrane symbols as in Figures 6 and 9. Modified from Metcalfe (1990, 1996, 1998) and partly after Smith *et al.* (1981), Audley-Charles (1988) and Audley-Charles *et al.* (1988). Present day outlines are for reference only. Distribution of land and sea for Chinese blocks principally from Wang, (1985). Land and sea distribution for Pangea/Gondwanaland compiled from Golongka *et al.* (1994), Smith *et al.* (1994); and for Australia from Struckmeyer & Totterdell (1990).

The collision of North and South China was nearly complete, with exhumation of ultra-high-pressure metamorphics along the Qinling-Dabie suture zone. Sediment derived from the North-South China collisional orogen poured into the Songpan Ganzi accretionary basin producing huge thicknesses of flysch turbidites (Nie *et al.*, 1995).

2.7 *Jurassic (205 - 141 Ma)*

Australia remained in low to moderate southern latitudes in the Jurassic. Rifting and separation of the Lhasa, West Burma, Sikuleh, Mangkalihat, and West Sulawesi terranes from NW Australia occurred progressively during the Late Triassic to Late Jurassic. The Ceno-Tethys ocean basin opened behind these terranes as they separated from Gondwanaland (Figure 13). Final welding of North China to Eurasia also took place in the Jurassic and the initial pre-breakup rifting of the main Gondwanaland supercontinent began at this time.

2.8 *Cretaceous (141 - 65 Ma)*

The Lhasa terrane collided and sutured to Eurasia in earliest Cretaceous times (Figure 13). Gondwanaland broke up and India drifted north, making initial contact with Eurasia at the end of the Cretaceous. This early contact between India and Eurasia is indicated by palaeomagnetic data from the ninetyeast Ridge (Klootwijk *et al.*, 1992) and also by Late Cretaceous Eurasian frogs occurring in India. The small West Burma and Sikuleh terranes accreted to Sibumasu during the Cretaceous. Australia began to separate from Antarctica at about 85 Ma (early Late Cretaceous) and drift northwards, but a connection with Antarctica via Tasmania still remained until about 45 Ma.

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REFERENCES

Archbold, N.W. 1996. Paleobiogeography of Australian Permian brachiopod faunas. In P. Copper & J. Jin (eds), *Brachiopods*. (Proceedings of the Third International Brachiopod Congress): 19-23. Rotterdam: A.A. Balkema.

Audley-Charles, M.G. 1988. Evolution of the southern margin of Tethys (North Australian region) from Early Permian to Late Cretaceous. In M.G. Audley-Charles & A. Hallam (eds), *Gondwana and Tethys*, Geological Society Special Publication No. 37: 79-100. Oxford: Oxford University Press.

Audley-Charles, M.G., Ballantine, P.D. & Hall, R. 1988. Mesozoic-Cenozoic rift-drift sequence of Asian fragments from Gondwanaland. *Tectonophysics* 155: 317-330.

Aung, A.K. 1995. New Middle Devonian (Eifelian) rugose corals from Myanmar. *Journal of Southeast Asian Earth Sciences* 11: 23-32.

Brock, G.A., Engelbretsen, M. & Cockle, P. 1999. The significance of changing biogeographic signals for the East Gondwana-China connection during the Cambrian. In I. Metcalfe (ed), *Where Worlds Collide: Faunal and floral migrations and evolution in SE Asia-Australasia, Abstracts:* 12. Asia Centre, University of New England.

Burrett, C. 1973. Ordovician biogeography and continental drift. *Palaeogeography, Palaeoclimatology, Palaeoecology* 13: 161-201.

Burrett, C. & Stait, B. 1985. South-East Asia as part of an Ordovician Gondwanaland - a palaeobiogeographic test of a tectonic hypothesis. *Earth and Planetary Science Letters* 75: 184-190.

Burrett, C., Long, J. & Stait, B. 1990. Early-Middle Palaeozoic biogeography of Asian terranes derived from Gondwana. In W.S. McKerrow & C.R. Scotese (eds), Palaeozoic Palaeogeography and Biogeography. *Geological Society Memoir* 12: 163-174.

Chen, Z.Q & Shi, G.R. 1999. *Chuiella* gen. nov. (Brachiopoda) and palaeoecology from the Lower Carboniferous of the Kunlun Mountains, NW China. *Alcheringa* 23: 259-275.

Cocks, L.R.M. & Fortey, R.A. 1997. A new *Hirnantia* fauna from Thailand and the biogeography of the latest Ordovician of South-East Asia. *GEOBIOS* 20: 117-126.

Fortey, R.A. 1989. An Early Devonian trilobite fauna from Thailand. *Alcheringa* 13: 257-267.

Fortey, R.A. 1997. Late Ordovician trilobites from southern Thailand. *Palaeontology* 40: 397-449.

Fortey, R.A. & Cocks, L.R.M. 1998. Biogeography and palaeogeography of the Sibumasu terrane in the Ordovician: a review. In R. Hall, & J.D. Holloway (eds), *Biogeography and Geological Evolution of SE Asia:* 43-56. , Amsterdam, The Netherlands: Backhuys Publishers.

Golonka, J., Ross, M.I. & Scotese, C.R. 1994. Phanerozoic paleogeographic and paleoclimatic modeling maps. In A.F. Embry, B. Beauchamp & D.J. Glass (eds), *Pangea: Global Environments and Resources*. Canadian Society of Petroleum Geologists, Memoir 17: 1-47.

Jell, P.A., Burrett, C.F., Stait, B. & Yochelson, E.L. 1984. The Early Ordovician bellerophontoid *Peelerophon oehlerti* (Bergeron) from Argentina, Australia and Thailand. *Alcheringa* 8: 169-176.

Johanson, Z. & Ritchie, A. 1999. Sarcopterygian fishes from the Hunter Siltstone (Late Famennian) near Grenfell, NSW, Australia. In I. Metcalfe (ed), *Where Worlds Collide: Faunal and floral migrations and evolution in SE Asia-Australasia, Abstracts:* 37. Asia Centre, University of New England.

Jones, P.J., Metcalfe, I., Engel, B. A., Playford, G., Rigby, J., Roberts, J., Turner, S. & Webb, G.E. 2000. Carboniferous Palaeobiogeography of Australasia. In T. Wright, G. Young, & J. Talent (eds) *Palaeobiogeography of Australasian Faunas and Floras,* Australasian Association of Palaeontologists Memoir 23: 259-286.

Klootwijk, C. 1996. Phanerozoic configurations of Greater Australia: Evolution of the North West Shelf. Part Two: Palaeomagnetic and geologic constraints on reconstructions. *AGSO Record* 1996/52: 85p.

Klootwijk, C.T., Gee, J.S., Peirce, J.W., Smith, G.M. & McFadden, P.L. 1992. An early India-Asia contact: Paleomagnetic constraints from Ninetyeast Ridge, ODP Leg 121. *Geology* 20: 395-398.

Laurie, J.R. & Burrett, C. 1992. Biogeographic significance of Ordovician brachiopods from Thailand and Malaysia. *Journal of Paleontology* 66: 16-23.

Laveine, J.P., Ratanasthein, B. & Azhar Haji Hussin. 1999. The Carboniferous floras of Southeast Asia: Implications for the relationships and timing of accretion of some Southeast Asian Blocks. In I. Metcalfe (ed) *Gondwana dispersion and Asian accretion*, Final Results Volume for IGCP Project 321: 229-246. Rotterdam: A.A. Balkema.

Li, Z.X., Zhang, L. & Powell, C.McA. 1995. South China in Rodinia: Part of the missing link between Australia-East Antarctica and Laurentia? *Geology* 23: 407-410.

Li, Z.X., Zhang, L. & Powell, C. McA. 1996. Positions of the East Asian cratons in the Neoproterozoic supercontinent Rodinia. *Australian Journal of Earth Sciences* 43: 593-604.

Lin Baoyu & Webby, B.D. 1989. Biogeographic relationships of Australian and Chinese Ordovician corals and stromatoporoids. *Memoirs of the Association of Australasian Palaeontologists* 8: 207-217.

Liu Benpei, Feng Qinglai & Fang Nianqiao. 1991. Tectonic evolution of the Palaeo-Tethys in Changning-Menglian Belt and adjacent regions, western Yunnan. *Journal of China University of Geosciences* 2: 18-28.

Liu Benpei, Feng Qinglai, Fang Nianqiao, Jia Jinhua, He Fuxiang, Yang Weiping & Liu Diansheng. 1996. Tectono-paleogeographic framework and evolution of the Paleotethyan archipeligoes ocean in Changning-Menglian belt, Western Yunnan, China. In Fang Nianqiao & Feng Qinglai (eds) *Devonian to Triassic Tethys in Western Yunnan, China*: 1-12. China University of Geosciences Press.

Long, J.A. 1993. Palaeozoic vertebrate biogeography of south-east Asia and Japan. In J.A. Long (ed) *Palaeozoic Vertebrate Biostratigraphy and Biogeography*: 277-289. London, Belhaven Press.

Long, J.A. & Burrett, C.F. 1989. Fish from the Upper Devonian of the Shan-Thai terrane indicate proximity to east Gondwana and South China terranes. *Geology* 17: 811-813.

Metcalfe, I. 1988. Origin and assembly of Southeast Asian continental terranes. In: M.G. Audley-Charles & A. Hallam (eds), *Gondwana and Tethys*. Geological Society of London Special Publication No. 37: 101-118.

Metcalfe, I. 1990. Allochthonous terrane processes in Southeast Asia. *Philosophical Transactions of the Royal Society of London* A331: 625-640.

Metcalfe, I. 1993. Southeast Asian terranes: Gondwanaland origins and evolution. In. R.H. Findlay, R. Unrug, M.R. Banks, & J.J. Veevers (eds) *Gondwana 8 - Assembly, Evolution, and Dispersal* (Proceedings Eighth Gondwana Symposium, Hobart, 1991): 181-200. Rotterdam: A.A. Balkema.

Metcalfe, I. 1996. Pre-Cretaceous evolution of SE Asian terranes In R. Hall & D. Blundell, (eds) *Tectonic Evolution of Southeast Asia*. Geological Society Special Publication No. 106: 97-122.

Metcalfe, I. 1998. Palaeozoic and Mesozoic geological evolution of the SE Asian region: multidisciplinary constraints and implications for biogeography. In R. Hall, & J.D. Holloway (eds) *Biogeography and Geological Evolution of SE Asia*: 25-41. Amsterdam, The Netherlands: Backhuys Publishers.

Metcalfe, I. 1999. Gondwana dispersion and Asian accretion: an overview. In I. Metcalfe (ed) *Gondwana dispersion and Asian accretion*, Final Results Volume for IGCP Project 321: 9-28. Rotterdam: A.A. Balkema.

Metcalfe, I. 2000. The Bentong-Raub Suture Zone. *Journal of Asian Earth Sciences*, 18 (Part 6): 691-712.

Nicoll, R.S. & Metcalfe, I. 1994. Late Cambrian to Early Silurian conodont endemism of the Sinian-Australian margin of Gondwanaland. In Wang Zhi-hao & Xu Fang-ming (eds) *First Asian Conodont Symposium, Nanjing, China, Abstracts of Papers*: 7.

Nicoll, R.S. & Metcalfe, I. 1998. Early and Middle Permian conodonts from the Canning and Southern Carnarvon Basins, Western Australia: Their implications for regional biogeography and palaeoclimatology. *Proceedings Royal Society of Victoria* 110: 419-461.

Nicoll, R.S. & Metcalfe, I. 1999. Cambrian to Permian conodont biogeography in East Asia-Australasia. In I. Metcalfe (ed), *Where Worlds Collide: Faunal and floral migrations and evolution in SE Asia-Australasia, Abstracts*: 62-64. Asia Centre, University of New England.

Nicoll, R.S. & Totterdell, J.M. 1990. Conodonts and the distribution in time and space of Ordovician sediments in Australia and adjacent areas. *Tenth Australian Geological Convention, Geological Society of Australia, Abstracts* 25: 46.

Nie, Y.S., Yin, A., Rowley, D.B.& Jin, Y. 1995. Exhumation of the Dabie Shan ultra-high pressure rocks and accumulation of the Songpan-Ganzi flysch sequence, central China. *Geology* 22: 999-1002.

Ritchie, A., Wang Shitao, Young, G.C. & Zhang Guorui. 1992. The Sinolepidae, a family of Antiarchs (Placoderm Fishes) from the Devonian of South China and eastern Australia. *Records of the Australian Museum* 44: 319-370.

Rong Jia-Yu, Boucot, A.J., Su Yang-Zheng & Strusz, D.L. 1995. Biogeographical analysis of Late Silurian brachiopod faunas, chiefly from Asia and Australia. *Lethaia* 28: 39-60.

Sengör, A.M.C., Altiner, D., Cin, A., Ustaomer, T. & Hsu, K.J. 1988. Origin and assembly of the Tethyside orogenic collage at the expense of Gondwana Land. In M.G. Audley-Charles & A. Hallam (eds) *Gondwana and Tethys*. Geological Society Special Publication No. 37: 119-181. Oxford: Oxford University Press.

Servais, T., Brocke, R. & Fatka, O. 1996. Variability in the Ordovician acritarch *Dicrodiacrodium*. *Palaeontology* 39: 389-405.

Shergold, J., Burrett, C., Akerman, T. & Stait, B. 1988. Late Cambrian trilobites from Tarutao Island, Thailand. *New Mexico Bureau of Mines & Mineral Resources Memoir* 44: 303-320.

Shi, G.R. & Archbold, N.W. 1998. Permian marine biogeography of SE Asia. In R. Hall & J.D. Holloway (eds) Biogeography and Geological Evolution of SE Asia: 57-72. Amsterdam, The Netherlands: Backhuys Publishers.

Singharajwarapan, S. & Berry, R. 2000. Tectonic implications of the Nan Suture Zone and its Relationship to the Sukhothai Fold Belt. *Journal of Asian Earth Sciences* 18 (Part 6): 663-673.

Smith, A.G., Hurley, A.M. & Briden, J.C. 1981. *Phanerozoic palaeocontinental world maps*. Cambridge: Cambridge University Press.

Smith, A.G, Smith, D.G. & Funnell, B.M. 1994. *Atlas of Mesozoic and Cenozoic coastlines*. Cambridge, Cambridge University Press.

Stait, B. & Burrett, C.F. 1982. *Wutinoceras* (Nautiloidea) from the Setul Limestone (Ordovician) of Malaysia. *Alcheringa* 6: 193-196.

Stait, B. & Burrett, C.F. 1984. Ordovician nautiloid faunas of Central and Southern Thailand. *Geological Magazine* 121: 115-124.

Stait, B., Wyatt, D. & Burrett, C.F. 1987. Ordovician nautiloid faunas of Langkawi Islands, Malaysia and Tarutao Island, Thailand. *N. Jb. Geol. Palaont. Abh.* 174: 373-391.

Struckmeyer, H.I.M. & Totterdell, J.M. (Coordinators) and BMR Palaeogeographic Group. 1990. *Australia: Evolution of a continent*. Canberra: Bureau of Mineral Resources, Australia.

Tong-Dzuy Thanh & Cai Chongyang. 1995. Devonian flora of Vietnam. In Tran Van Tri (ed) Proceedings of the International Symposium Geology of Southeast Asia and adjacent areas. *Journal of Geology*, Series B (Geological Survey of Vietnam), No 5-6: 105-113.

Tong-Dzuy Thanh, Janvier, P. & Ta Hoa Phuong. 1996. Fish suggests continental connections between the Indochina and South China blocks in middle Devonian time. *Geology* 24: 571-574.

Ueno, K. 1999. Gondwana/Tethys divide in east Asia: solution from Late Paleozoic foraminiferal paleobiogeography. In Ratanasthein, B. & Rieb, S.L. (eds) *Proceedings of the International Symposium on Shallow Tethys (ST) 5, Chiang Mai, Thailand, February, 1999*: 45-54. Faculty of Science, Chiang Mai University.

Wang, H. 1985. *Atlas of the palaeogeography of China*. Beijing, Cartographic Publishing House.

Webby, B.D., Wyatt, D. & Burrett, C. 1985. Ordovician stromatoporoids from the Langkawi Islands, Malaysia. *Alcheringa* 9: 159-166.

Yang Jialu. 1994. Cambrian. In Yin Hongfu (ed). *The palaeobiogeography of China*. Oxford Biogeography Series No.8: 35-63. Oxford, Clarendon Press.

Young, G.C. 1981. Biogeography of Devonian vertebrates. *Alcheringa* 5: 225-243.
Young, G.C. 1987. Devonian palaeontological data and the Armorica problem. *Palaeogeography, Palaeoclimatology, Palaeoecology* 60: 283-304.
Young, G.C. & Janvier, P. 1999. Early-middle Palaeozoic vertebrate faunas in relation to Gondwana dispersion and Asian accretion. In I. Metcalfe (ed) *Gondwana dispersion and Asian accretion*, Final Results Volume for IGCP Project 321: 115-140. Rotterdam: A.A. Balkema.
Zhao, X., Coe, R.S., Gilder, S.A. & Frost, G.M. 1996. Palaeomagnetic constraints on the palaeogeography of China: implications for Gondwanaland. *Australian Journal of Earth Sciences* 43: 643-672.
Zhou, Z., Dean, W.T & Luo, H. 1998a. Early Ordovician trilobites from Dali, West Yunnan, China, and their palaeogeographical significance. *Palaeontology* 41: 429-460.
Zhou, Z., Dean, W.T, Yuan, W. & Zhou, T. 1998b. Ordovician trilobites from the Dawangou Formation, Kalpin, Xinjiang, North-West China. *Palaeontology* 41: 693-735.